Title: With a suitable example demonstrate the perceptron learning law with its decision regions using python.Give the output in graphical form

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]])
y = np.array([0, 0, 0, 1])

model = Perceptron(max_iter=1000, eta0=0.1, random_state=42)
model.fit(X, y)

y_pred = model.predict(X)
accuracy = accuracy_score(y, y_pred)

print("Actual Output:", y)
print("Predicted Output:", y_pred)
print("Accuracy:", accuracy * 100, "%")
```

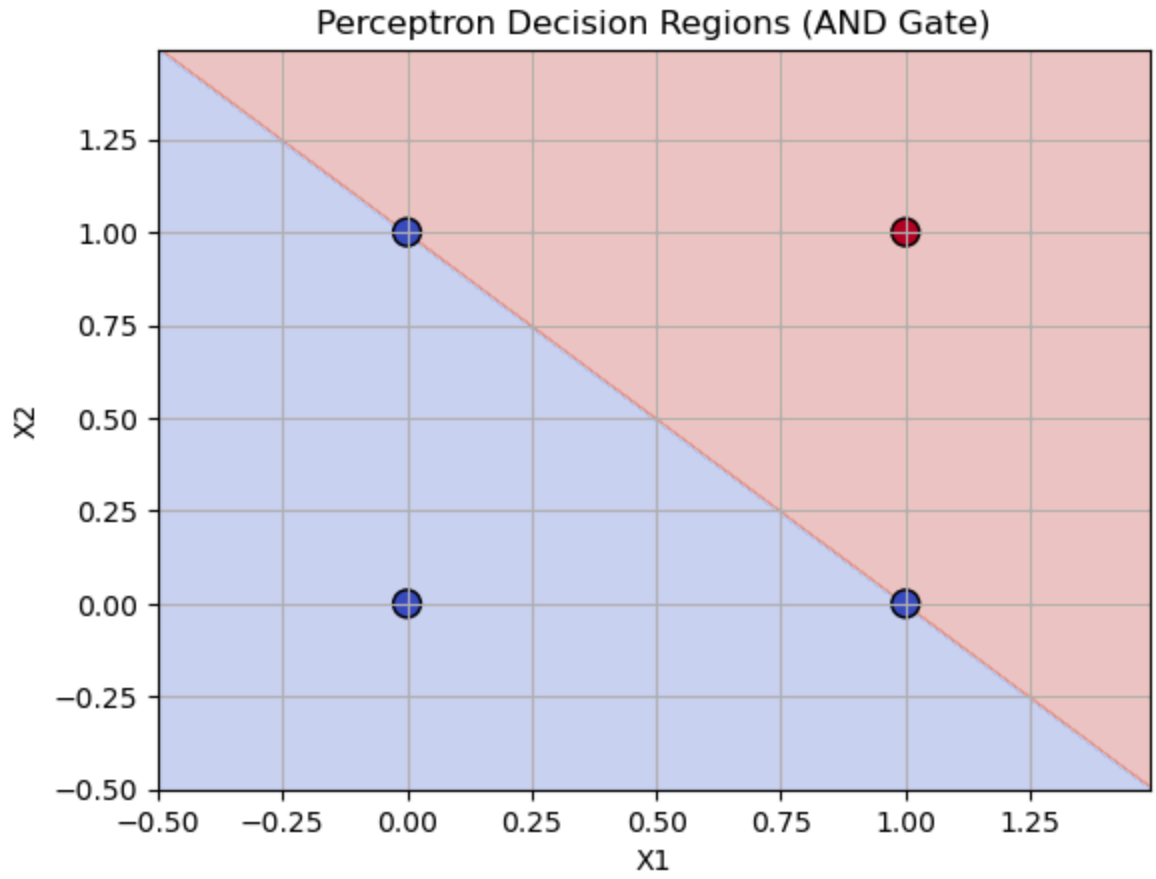
```
Actual Output: [0 0 0 1]
Predicted Output: [0 0 0 1]
Accuracy: 100.0 %
```

In [3]:
```
x1, x2 = np.meshgrid(
    np.arange(-0.5, 1.5, 0.01),
    np.arange(-0.5, 1.5, 0.01)
)

Z = model.predict(np.c_[x1.ravel(), x2.ravel()])
Z = Z.reshape(x1.shape)

plt.contourf(x1, x2, Z, alpha=0.3, cmap="coolwarm")
plt.scatter(X[:, 0], X[:, 1], c=y, cmap="coolwarm",
            edgecolors="black", s=100)

plt.xlabel("X1")
plt.ylabel("X2")
plt.title("Perceptron Decision Regions (AND Gate)")
plt.grid()
plt.show()
```


Perceptron Decision Regions (AND Gate)
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
X2
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
X1

In [ ]: